Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C++/Python systems, quantitative applications, and distributed services**; strong in algorithms, data structures, concurrency, asynchronous processing, performance optimization, mathematical modeling, and collaborative delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Engineered **concurrent distributed Python services** processing 1,000+ asynchronous jobs with explicit states, failure isolation, and optimized data access, increasing throughput 3x while preventing cascading failures under load.
- Optimized PostgreSQL and SQL Server workloads across 10,000+ records using indexing, validation, and performance analysis, reducing latency 35% while sustaining 99%+ reliability across 5,000+ monthly production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built a **parallel Python processing engine** using ThreadPoolExecutor, secure file ingestion, and fault-tolerant execution, handling 1,000+ files per batch and 10,000+ daily requests with stable peak-load performance.
- Owned features across design, implementation, testing, code review, deployment, debugging, and support, collaborating with technical and non-technical stakeholders while reducing feedback turnaround 60% through practical workflow automation.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Translated **quantitative models into efficient Python systems** for autonomous-driving research, building data and simulation pipelines over 100,000+ trajectories and improving long-horizon planning performance 18% across controlled evaluations.
- Designed a spatial-temporal model combining road geometry, lane direction, and historical trajectories through multi-head attention, improving closed-loop stability 15% while evaluating quantitative metrics across 10,000 driving episodes.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **Java and Python application components** using OOP design patterns, SQL, caching, and API integrations across Linux, Windows, and macOS, reducing response times 30% and technical debt 20%.
- Built reliable development workflows with Git, Jenkins, PyTest, and Mockito, executing 20+ tests and parallel builds while collaborating with Microsoft and GitHub engineers across globally distributed Agile teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **scalable high-performance financial applications** for pricing, payments, refunds, transaction status, and reporting, supporting 50,000+ production records with real-time synchronization and sub-200-millisecond API responses across platforms.
- Gathered customer requirements and documented 12+ system interfaces, implementing secure transaction workflows, analytics pipelines, and webhook automation while reducing manual refund handling 70% across mobile and web applications.

Skills

Languages: C++, Python, Java, SQL, Bash, Shell, JavaScript, TypeScript
Computer Science: Data structures, algorithms, OOP, design patterns, concurrency, multithreading, parallel processing, asynchronous architectures
Systems: Linux, distributed systems, networking, TCP/IP, performance optimization, caching, REST APIs, WebSockets
Quantitative: Mathematical modeling, data capture, data analysis, simulation, transactions, pricing, financial applications
Engineering: Git, Jenkins, GitHub Actions, CI/CD, Docker, Kubernetes, PyTest, debugging, code reviews, Agile collaboration

Projects

CPU Scheduling Simulator Live | *C++, React.js, Operating Systems*

- Implemented and benchmarked **six C++ scheduling algorithms** with interactive controls, Gantt visualization, comprehensive edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25%.

FinNexus Live | *Kubernetes, MongoDB, WebSockets*

- Built a **distributed financial platform** supporting 50,000+ users and 1M+ transactions with secure bank integrations, real-time updates, MongoDB persistence, advanced filtering, and Kubernetes-based scalable deployment automation.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Enhanced a **cross-platform database application** across Java and TypeScript components, supporting 10+ SQL and NoSQL systems with structured execution, validation, Docker, Maven, Linux workflows, and reusable interfaces.